

# Turtle Bricks — Pollinator Alley

FLL BIOGLOW 2026-27 · Innovation Project summary · Oct 4, 2026

Pollinator Alley is our Turtle Bricks Innovation Project for FLL BIOGLOW 2026-27. This is our work from our Research & Ideas Sheets, cleaned up and checked against sources.

**Verification needed.** Three facts in this doc did not come from the team — the coach added them, and they're marked **Verification needed** below. Before the team uses one, a team member should open the source with a grown-up, find the fact, and say it in their own words.

## The problem

**How do we keep pollinators alive and safe?**

## Why it matters

Pollinators help plants make fruit and seeds. About one third of what farms grow needs pollinators (USDA). More than 3 out of 4 kinds of food crops need them at least a little (IPBES, a worldwide group of scientists). Without pollinators we would lose many fruits, nuts and vegetables. Some plants used for cloth, cooking oil and medicine need pollinators too (USDA, same page).

## What we learned

| FACT                                                                                                                                                                              | SOURCE                  | STATUS                           |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|----------------------------------|
| Pollinators carry pollen from flower to flower. This is how many plants make fruit and seeds.                                                                                     | IPBES                   | ✓ We found it, source checked    |
| There are about 20,000 kinds of bees in the world. About 3,500 live in the United States.                                                                                         | U. of Minnesota Bee Lab | ✓ We found it, source checked    |
| Pollinators are hurt when people cut down plants and flowers, use pesticides, and change the climate.                                                                             | IPBES                   | ✓ We found it, source checked    |
| Big doses of some pesticides can kill pollinators.                                                                                                                                | IPBES                   | ✓ We found it, source checked    |
| A healthy bee colony loses 800 to 1,200 bees every day in summer. This is normal — worker bees live only 4 to 6 weeks, and the queen lays about 2,000 eggs a day to replace them. | Honey Bee Suite         | ✓ We found it, meaning corrected |

| FACT                                                                                                                                                                     | SOURCE       | STATUS                              |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-------------------------------------|
| More than 4 out of 10 kinds of bees, butterflies and other small pollinators are in danger of dying out.                                                                 | IPBES        | ⚠ Verification needed (coach added) |
| Cars kill bees. Scientists in Utah took 29 car trips, and every trip killed at least one bee. They think cars kill tens of millions of bees each day in the western U.S. | Science News | ⚠ Verification needed (coach added) |
| Scientists already use smart cameras to count pollinators. On one strawberry farm, nine small cameras could tell the pollinators apart and count their flower visits.    | Cosmos       | ⚠ Verification needed (coach added) |

## Our solution: Pollinator Alley

Pollinator Alley is a stretch of land set aside for pollinators. It has many flowers and no buildings, and the roads through it are covered so that traffic does not harm the pollinators.

| PART          | WHAT IT DOES                                                                                 | WHY WE THINK IT WILL HELP                                                             |
|---------------|----------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| Covered roads | Keep pollinators and cars apart                                                              | Cars kill lots of bees (verification needed — see the Utah fact above)                |
| A big garden  | Gives pollinators a place to live and eat                                                    | Pollinators are hurt when plants and flowers are cut down                             |
| No pesticides | Keeps poison out of the alley                                                                | Some pesticides can kill pollinators                                                  |
| AI cameras    | Take photos of pollinators, check how they are doing, and send the information to scientists | Scientists already do this on farms (verification needed — see the camera fact above) |

## Where it could go: Data Center Alley

*This section is a coach's idea for the team to react to. Every fact in it is marked verification needed until a team member checks it.*

**The idea in one sentence.** Put Pollinator Alley inside Data Center Alley: turn the grass around the data centers and under the power lines into one long path of flowers.

### Why here

Data Center Alley is in Loudoun County, Virginia. It has about 200 data centers, close to 50 million square feet of buildings (Lightyear guide, June 2026). *Verification needed.*

Data centers have lots of mowed grass, fences and power lines around them. Mowed grass has almost no flowers, so pollinators find little to eat there.

One new data center plan in Loudoun promises to keep 60% of its land as open space, plus a wide strip of trees (Loudoun Now). The article says nothing about flowers or pollinators — that open space is our chance. *Verification needed.*

## How the team's ideas fit

| TEAM IDEA                       | IN DATA CENTER ALLEY                                                                                                       |
|---------------------------------|----------------------------------------------------------------------------------------------------------------------------|
| Many flowers and zero buildings | The green strips between the data centers become flower meadows. No buildings go on the strips.                            |
| A big garden                    | Each data center swaps some of its lawn for a native flower garden.                                                        |
| No pesticides                   | The data centers agree not to spray the meadows.                                                                           |
| Covered roads                   | Where the flower path crosses a road, a covered crossing keeps pollinators away from cars.                                 |
| Pollinator AI data center       | Cameras in the meadows send pollinator photos to the data center next door. The buildings that run AI help count the bees. |

**Someone already does part of this.** Dominion Energy stopped mowing everything under some of its power lines in Virginia so native flowers can grow back for pollinators (Dominion Energy, 2017). *Verification needed.*

**What about on top of the data centers?** Flower gardens on the roofs would add habitat without using any new land. Each fact below is *verification needed*:

- It has been done — a data center in Vienna, Austria has a green roof with four beehives on it (Data Center Dynamics, 2022).
- Bees do use roof gardens, but the higher the roof, the fewer bees; scientists found roofs above the eighth floor do not attract bees (U. of Toronto, 2017). Most data centers are only a few floors tall.
- Roofs with many kinds of native flowers help native bees more than roofs with one kind of plant (U. of Toronto, same page).
- Data center workers in Loudoun have already helped bees — in 2022, 145 volunteers from the data center industry built pollinator garden beds at JK Community Farm (Data Center Knowledge).

**Questions to ask a data center engineer:** Data center roofs hold big cooling machines that blow hot air. Is there room for a garden? Can the roof hold the weight of wet soil?

**A test the team can run.** Pick a patch of mowed grass and a patch of flowers. Watch each one for 10 minutes and make a tally mark for every pollinator that lands. Do it on three different days. The two totals show why lawns should become meadows.

**A model the team can build.** LEGO data centers on a baseplate with a green flower path winding between them, a covered road crossing, and a camera on a post.

## People to ask

- **Loudoun Wildlife Conservancy** — runs a Protecting Pollinators program and sells native plants. Email: [contact@loudounwildlife.org](mailto:contact@loudounwildlife.org)
- **A data center company:** "Would you plant a pollinator meadow on your land?"
- **Dominion Energy:** "Which flowers grow back under your power lines?"

**One thing for the team to decide.** The original idea said zero buildings. Data Center Alley is full of buildings. The team can say the flower path itself has zero buildings, or pick a different place.

## How we will test it

- Build a miniature model of Pollinator Alley.
- Plant a flower bed and count which pollinators visit.
- Build a LEGO data center with a camera that collects data and sends it.

## How we will share it

We will show our model so people can see what Pollinator Alley could look like, post it online, and send our data to scientists.

## Questions we still have

- What usually kills pollinators?
- Why do people use pesticides?
- Should the alley have trees as well as flowers?
- How else could AI help pollinators?

## What changed from our sheets

- "One third of food" became "about one third of what farms grow needs pollinators." Wheat, corn and rice do not need pollinators, and some writers say the one-third number is too big (Genetic Literacy Project).
- "More than 20,000 kinds of bees" became "about 20,000."
- The 800–1,200 bees a day number is real, but it is about a healthy colony — it does not show that bees are in trouble.
- "Put bees in the flower bed" became "count which pollinators visit."

**Verification needed.** The coach added these three facts; the team has not checked them yet: (1) many kinds of pollinators are in danger of dying out; (2) cars kill a lot of bees; (3) scientists already use smart cameras to count pollinators on farms.

## Sources

*Pages opened and checked on October 4, 2026.*

- USDA — Hill Farm Buzzing with Pollinator Success
- Future Earth — IPBES releases global pollinators assessment
- University of Minnesota Bee Lab — Native bee diversity
- Science News — Bees flying near cars are dying by the millions, a roadkill study suggests
- Honey Bee Suite — Counting dead bees (a beekeeping website, not a scientific study)
- Cosmos — Artificial intelligence eye on pollinators
- Genetic Literacy Project — critique of the "one third" claim